eSynth Lost Sampler

Schematic-reviewed proposed hardware correction for 10 seconds of mono audio at 12-bit / 50 kHz

PUBLICATION STATUS	Ready to publish as a proposed solution. Not a verified working build.

Executive summary

The attached seven-page ESynth sampler document defines a target of 12-bit mono sampling at 50 kHz for 10 seconds with a PIC16F1784. Its own text reports that the original I2C EEPROM approach did not work. Inspection of the attached schematic and listing reveals additional gaps: the memory capacity is described using incorrect units, the displayed schematic does not show the replacement SPI memory, the existing design uses a 5 V rail, and the listed firmware does not implement a complete record-and-playback data path.

Recommended architecture

Audio input	PIC16F1784	Level-compatible SPI	CY15B108QN F-RAM
Bias + anti-alias filter verified against the ADC range	ADC at 50 ksps; 12-bit sample packing; timer-driven capture	SCK, MOSI, MISO, CS; proper 5 V / 3.3 V interface	8 Mbit = 1,048,576 bytes; non-volatile, no EEPROM page-write wait

What this document does and does not claim

It records the corrections justified by the supplied schematic, code listing, and manufacturer datasheets; specifies a feasible memory/interface architecture; and provides a release checklist. It does not claim that a revised schematic has been built, that firmware has been compiled, or that recording/playback has passed bench tests.

A percentage such as “98% working” cannot be supported without a successful build and hardware test. The defensible status is “proposed solution ready for review/publication”.

1. Findings from the supplied schematic and code

The following observations refer to the user-provided file “ESynth sampler.pdf”, especially its sampler description (printed page 43), circuit drawing and 5 V regulator (printed page 44), firmware description (printed page 45), and mikroC listing (printed pages 46–49).

Item	What the attachment shows	Engineering consequence
Original memory units	The description calls the 24LC1025 a “1 MByte” EEPROM. The manufacturer specifies the 24AA/LC/FC1025 family as 1024 Kbit = 128K × 8 = 131,072 bytes.	Correct the unit: 1 Mbit is not 1 MByte. One 24LC1025 cannot hold a 10-second recording at the target settings.
Two 24LC512 devices	The narrative says two 24LC512 devices were tried. Each is 512 Kbit = 65,536 bytes; two total 131,072 bytes. The shown circuit still draws a connected U2 24AA1025, while U3/U4 24LC512 symbols are shown separately and are not drawn as a completed replacement wiring.	The attachment is not an as-built schematic for a dual-EEPROM or SPI-F-RAM revision. The actual netlist/addressing must be confirmed.
Existing supply	The power drawing identifies an LP2957 5 V output. The proposed CY15B108QN operates at 1.8–3.6 V.	Do not connect the F-RAM directly to 5 V PIC outputs. Add suitable level translation or redesign and revalidate the supply domain.
Firmware data format	The helper routine shown in the listing writes ADRESH and ADRESL as two separate bytes. The interrupt also stores ADRESH/ADRESL in separate 128-element arrays.	The 750,000-byte packed format in this proposal requires a new, tested pack/unpack path. The listing is not a drop-in firmware update.
Record/playback flow	The displayed listing does not show a complete, continuously scheduled flush of every sample buffer to memory or a complete streaming playback routine. The source itself describes a non-working implementation.	Implement and bench-test buffer handoff, address management, sample count, end-of-recording, and playback timing.
Clock/timing	The schematic shows X1 and the timer code claims a 20 µs period, but the crystal frequency is not labelled in the drawing.	The 50 kHz timer reload and ADC acquisition timing cannot be certified from the attachment alone.

Important source distinction

The original text gives the intended product behaviour, but the provided source listing is evidence of an incomplete implementation, not proof that a particular firmware flow works. In particular, the memory technology change alone cannot complete recording and playback.

Capacity sanity check

At 50,000 samples/s for 10 s, the target contains 500,000 samples. Saving each 12-bit sample in a 16-bit slot needs 1,000,000 bytes. Packing two 12-bit samples into three bytes needs 750,000 bytes. Both are much larger than 131,072 bytes.

2. Storage sizing and selected memory

The proposed replacement is a single Infineon CY15B108QN-40SXI SPI F-RAM, organized as 1,048,576 × 8 bytes (8 Mbit), rated up to 40 MHz SPI and 1.8–3.6 V. Its direct-write behaviour removes the EEPROM page-program delay; the PIC firmware still needs buffering and correct timing.

Quantity	Calculation	Result
Samples in 10 s	50,000 samples/s × 10 s	**500,000 samples**
Packed 12-bit data	500,000 × 12 / 8	**750,000 bytes**
16-bit storage slots	500,000 × 2	**1,000,000 bytes**
Two original 24LC512	2 × 512 Kbit / 8	**131,072 bytes**
One original 24LC1025	1024 Kbit / 8	**131,072 bytes**
Proposed F-RAM	8 Mbit / 8	**1,048,576 bytes**

Capacity after a 256-byte metadata/header reservation

Storage format	Audio + 256-byte header	Free F-RAM space
Packed 12-bit	750,256 bytes	**298,320 bytes**
Two bytes/sample	1,000,256 bytes	**48,320 bytes**

Recommended stored format

Use a documented packed stream if the 750,000-byte layout is desired: for samples a and b, byte 0 = a[11:4]; byte 1 = a[3:0] in the high nibble plus b[11:8] in the low nibble; byte 2 = b[7:0]. Packing state must persist across buffer boundaries. Alternatively, storing each sample in two bytes is simpler and still fits the 1 MiB F-RAM with a 256-byte header, but gives less free space.

Illustrative memory map

0x00000–0x000FF: 256-byte header/metadata. Packed audio begins at 0x00100 and occupies 750,000 bytes through 0xB72AF. Keep a version, sample rate, sample count, format identifier, checksum and valid marker; write the valid marker only after the final buffer has been flushed.

3. Required schematic changes

The F-RAM is not a pin-compatible replacement for the I2C EEPROM. The new schematic must explicitly show the SPI nets, supply and level translation. The original drawing can be used as a reference for the controller, LCD, buttons and audio interface, but it is not the finished schematic for this revision.

F-RAM signal	PIC / board-side connection	Requirement
SCK	PIC MSSP SPI SCK (candidate signal: RC3/SCK)	Translate the 5 V PIC output to the F-RAM logic rail if PIC remains at 5 V.
SI / MOSI	PIC MSSP SPI SDO (candidate signal: RC5/SDO)	Translate 5 V to 3.3 V when required by the selected rail strategy.
SO / MISO	PIC MSSP SPI SDI (candidate signal: RC4/SDI)	Verify the input-high threshold; use a suitable 3.3 V-to-PIC interface if the PIC runs at 5 V.
CS	One free digital output, selected after netlist review	Dedicated chip-select; hold inactive high during reset and before initialization.
WP, HOLD	Tie to inactive levels as specified by the exact F-RAM datasheet	Do not leave control pins floating.
VDD, VSS	Regulated 3.3 V-compatible supply and common ground	Place 100 nF decoupling capacitor close to the F-RAM supply pins.

Power and logic levels

If retaining the existing 5 V regulator, specify and draw level translation for SCK, MOSI, CS and MISO. Do not connect 5 V PIC outputs directly to the 1.8–3.6 V F-RAM. A different approach is to operate the PIC at a validated 3.3 V rail, but this requires rechecking the oscillator, LCD, audio path, LEDs, and every connected device against their voltage limits.

Clock and analogue path

Specify the actual X1 frequency (or explicitly select the PIC internal oscillator), then recalculate the Timer1 reload and verify the 20 µs sample interval. Confirm that the analogue input is biased within the ADC input range and has appropriate anti-alias filtering. Confirm DAC output filtering and audio jack levels as part of playback validation.

Playback resolution

The PIC16F1784 includes an 8-bit DAC. Storing 12-bit samples does not give 12-bit analogue playback by itself. For playback through the internal DAC, scale samples to 8 bits; if 12-bit analogue output is a requirement, add an external 12-bit DAC and schedule memory reads so they cannot miss output deadlines.

PIC RAM budget

The PIC16F1784 has 512 bytes of SRAM. Two 128-byte buffers consume 256 bytes, leaving 256 bytes for all other variables and library/runtime needs. Do not retain the original 128-byte ADRESH and 128-byte ADRESL arrays in addition to the new double buffers without checking the compiler memory map. If SRAM is tight, reduce buffer size and re-evaluate the timing budget.

4. Firmware plan and release checklist

The revised firmware should be implemented as a buffered data logger/player. Keep the 50 kHz sample interrupt short: acquire one ADC sample, pack/store it in the active RAM buffer, and return. Never run a full SPI write from the time-critical ADC interrupt.

Recording sequence

1. Initialise oscillator, ADC, sample timer, SPI, chip-select, buffers and F-RAM state. Read a device/status value where supported; verify the memory responds before arming recording.
2. Invalidate the recording marker. Acquire exactly 500,000 samples at the verified 20 µs cadence. Pack or store samples using one specified format.
3. Use two RAM buffers. On completion of one buffer, switch immediately to the other; foreground code transmits the completed buffer to F-RAM. Confirm worst-case interrupt latency, overrun handling and buffer hand-off.
4. Flush the final partial buffer. Write metadata, sample count and checksum, then set the valid marker last. If the recording is interrupted, metadata must not mark it valid.
5. For playback, use read-ahead buffers and a deterministic output timer. Do not perform blocking 128-byte reads inside the audio-output interrupt.

Throughput estimate

Check	Value	Meaning
Sampling interval	**20 µs**	50,000 samples/s
Packed input rate	**75,000 bytes/s**	12 bits × 50,000 / 8
128-byte buffer fill	**1.7067 ms**	128 / 75,000 s
SPI write burst at 8 MHz	**About 133 µs**	WREN + WRITE opcode + 3-byte address + 128 data bytes; excludes software overhead
Nominal SPI bus share	**About 7.8%**	Does not prove real-time correctness; measure the compiled firmware.

Release acceptance checklist

[ ]	Final schematic shows F-RAM, all SPI nets, chip-select, voltage translation and decoupling.
[ ]	X1 / oscillator frequency and timer reload are explicitly documented.
[ ]	mikroC build completes cleanly; archive toolchain version and configuration words.
[ ]	RAM map confirms enough space for buffers and all variables.
[ ]	Logic analyser confirms 20 µs sampling without missed conversions or buffer overruns.
[ ]	Record a known waveform; read back the exact sample count and pass checksum.
[ ]	Verify replay timing and output level; measure distortion/noise as required.
[ ]	Complete at least 10 record/read/play cycles, including power cycles and interrupted recording.

References

[1] Supplied attachment: “ESynth sampler.pdf”, pages 43–49 (sampler description, circuit diagram, code listing).
[2] Microchip, PIC16(L)F1784/6/7 datasheet DS41637: https://ww1.microchip.com/downloads/en/DeviceDoc/41637A.pdf
[3] Microchip, 24AA1025/24LC1025/24FC1025 datasheet DS21941: https://ww1.microchip.com/downloads/en/DeviceDoc/21941H.pdf
[4] Microchip, 24AA512/24LC512/24FC512 datasheet DS20001754: https://www.microchip.com/en-us/product/24FC512
[5] Infineon, CY15B108QN/CY15V108QN 8-Mbit SPI F-RAM datasheet: https://www.infineon.com/part/CY15B108QN-40SXI
[6] Upstream repository: https://github.com/fungousrex/esynth-lost-sampler/blob/main/README.md

Final status: ready to submit for review as a proposed solution. Do not describe the electronics or firmware as verified until the checklist passes.